

ADAPTIVE COGNITIVE-LOAD STUDY TIMER FOR PERSONALIZED BREAK SCHEDULING

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Project Proposal Report

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TABLE OF CONTENTS

Declaration	3
Abstract	4
List of Figures	4
List of Tables.....	5
List of Abbreviations.....	5
1 Introduction	5
2 Background and Literature Survey	7
2.1 Cognitive load and mental fatigue	7
2.2 Ecological momentary assessment and micro-EMA	7
2.3 Passive sensing via keystroke and touch dynamics	8
2.4 Privacy considerations	8
2.5 Current study tools and gaps	9
3 Research Gap and Problem Statement	11
3.1 Research Gap and Problem Statement	11
3.2 Problem Statement	12
4 Objectives.....	13
4.1 Main Objective.....	13
4.2 Specific Objectives.....	13
5 Methodology	14
5.1 Research Design.....	14
5.2 Data Collection.....	14
5.3 Feature Extraction and Modelling.....	16
5.4 Requirement specification.....	17
5.5 Implementation	19
5.6 Evaluation Plan	20
6 Description of personal and Facilities.....	21
Personal Roles and Expertise	21
Facilities and Equipment.....	22
7 Work Breakdown Structure (WBS)	22
8 Self-Evaluation Plan.....	23
9 Budget	24
10 Commercialization Potential.....	24
11 Conclusion	25
12 References	26

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate dissertation under my supervision.

Signature of the supervisor:

Date:

Abstract

Fixed-schedule study timers, such as the Pomodoro technique, assume that all learners have the same cognitive endurance and do not monitor the user's mental state. Research shows that mental workload fluctuates during sustained study sessions; typing performance tends to slow and errors rise as fatigue builds[1]. Experience-sampling methods like ecological momentary assessment (EMA) collect self-reports in-situ, but traditional multi-item scales are burdensome, leading to poor adherence [2]. Micro-EMA – single-question prompts delivered through brief taps – allows more frequent sampling and reduces participant burden [2], while passive computer interaction data such as keystroke/touch timing provide near-continuous, objective indicators of cognitive state [3].

This proposal outlines a novel sensing subsystem for an adaptive study timer. The subsystem fuses passive interaction cadence (inter-keystroke/touch intervals, pause bursts) with micro-EMA responses to produce a privacy-preserving, real-time estimate of cognitive load. By capturing only timing metadata, not the content of what is typed, the approach remains unobtrusive and respects user privacy [3]. The resulting load estimate will drive personalized break scheduling and other features in the overall application. This proposal details the background, identifies the research gap, states objectives, and describes the methodology, resources and evaluation plan to deliver and test the cognitive load estimator.

Keywords: cognitive load, typing dynamics, micro-EMA, computer interaction, personalized learning

List of Figures

1. **Figure 1.0:** Conceptual architecture of the study timer
2. **Figure 2.1:** Conceptual architecture of the dual-signal sensing subsystem
3. **Figure 2.2:** Comparison of EMA and passive sensing on burden and sampling frequency

List of Tables

1. **Table 5.1:** Summary of features used for cognitive load estimation
2. **Table 5.2:** Evaluation of metrics and targets
3. **Table 7.1:** Work breakdown structure (WBS)

List of Abbreviations

Abbreviation	Description
EMA	Ecological Momentary Assessment
IKI	Inter-keystroke interval
NASA-TLX	NASA Task Load Index
SDK	Software Development Kit
WBS	Work Breakdown Structure
UR	User requirements
FR	Functional requirements
NFR	Non functional requirements

1 Introduction

Studying effectively requires balancing deep focus with timely breaks. Popular tools like Pomodoro timers rigidly enforce a 25/5-minute cycle regardless of a learner's current state. This one-size-fits-all approach ignores the fact that mental workload waxes and wanes, leading to either premature interruptions or prolonged exhaustion. Modern computer devices offer an opportunity to personalise study schedules by sensing when cognitive load increases and suggesting breaks accordingly. Our project, Adaptive Cognitive-Load Study Timer, integrates four modules: **1 - Cognitive load estimator**, **2 - Adaptive and personalized break scheduler**, **3 - Chronotype personalized predictive heatmap**, and **4 - Intent-Lock overlay**. The overall system is an application that collects interaction cadence and brief self-reports, fuses them into a within-person load score, uses a contextual bandit to determine work/break intervals, gently discourages distractions through an

intent-lock overlay, and surfaces personal peak-focus periods through a chronotype heatmap.

This proposal concentrates on the first feature – the sensing subsystem. The goal is to design and evaluate a dual-signal method that passively collects computer interaction such as keystroke timings, keystroke continuousness, mistakes (backspace key is tracked), auto-correction event fires and pointer behavior and intermittently prompts a one-tap micro-EMA. By doing so, we aim to estimate cognitive load in real time with higher accuracy expectations with respect to currently available models while respecting user privacy. The sensing module is foundational: accurate load estimation enables the adaptive scheduler to time breaks effectively and informs the intent-lock and chronotype features. Without a reliable sensing layer, the higher-level adaptations would be blind guesses.

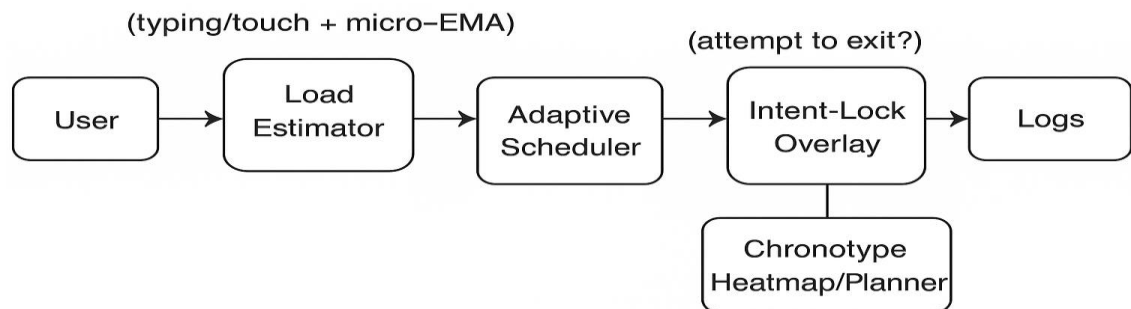


Figure 1.0: *Conceptual architecture of the study timer*

2 Background and Literature Survey

2.1 Cognitive load and mental fatigue

Cognitive load refers to the amount of mental effort required to perform a task. During prolonged work, mental fatigue builds up, manifesting as slower responses, lapses and increased errors [1]. De Jong et al.'s field study of office workers showed that typing speed declined and backspace corrections increased over the day, suggesting that mental fatigue can be detected through changes in typing dynamics [1]. Similar patterns have been reported in controlled experiments; participants initially maintain typing speed at the cost of accuracy, then gradually slow down as fatigue worsens [1]. Understanding when cognitive load rises is therefore crucial for timing restorative breaks.

Subjective workload scales, such as NASA's Task Load Index (NASA-TLX), have long been used to measure perceived workload. NASA-TLX is a widely used multidimensional self-report scale developed by NASA's Ames Research Center and cited in thousands of studies [5]. While reliable, these paper-and-pencil questionnaires are too intrusive for repeated in-the-moment assessments and require considerable effort to administer.

2.2 Ecological momentary assessment and micro-EMA

Ecological momentary assessment (EMA) prompts participants multiple times per day to report their thoughts, feelings and behaviors, reducing recall bias compared to retrospective questionnaires [2]. However, traditional EMA uses multi-item scales that remain burdensome and can reduce compliance [2]. To address this, researchers have developed micro-EMA – single-question prompts delivered through a quick tap on a smartphone or computer. Micro-EMA allows more frequent sampling with reduced burden and has shown higher adherence rates than multi-question EMAs [2]. The micro-Stress EMA framework proposed by King et al. demonstrated that reducing an EMA to one question makes it possible to detect subtle physiological changes and increases the frequency of self-reporting [2]. In

other words, micro-EMA captures momentary perceptions without overloading the participant.

2.3 Passive sensing via keystroke and touch dynamics

Computers have become ubiquitous, offering sensors that passively record user interactions. Nguyen et al. note that computers can collect passive data, such as keystroke dynamics, nearly continuously and unobtrusively, providing authentic, real-world measures of cognitive states [3]. Keystroke dynamics – metadata about keypress timing and other keyboard events – do not record the actual text, preserving privacy [3]. Passive data capture can complement self-reports, offering objective, high-frequency signals without conscious user effort [3].

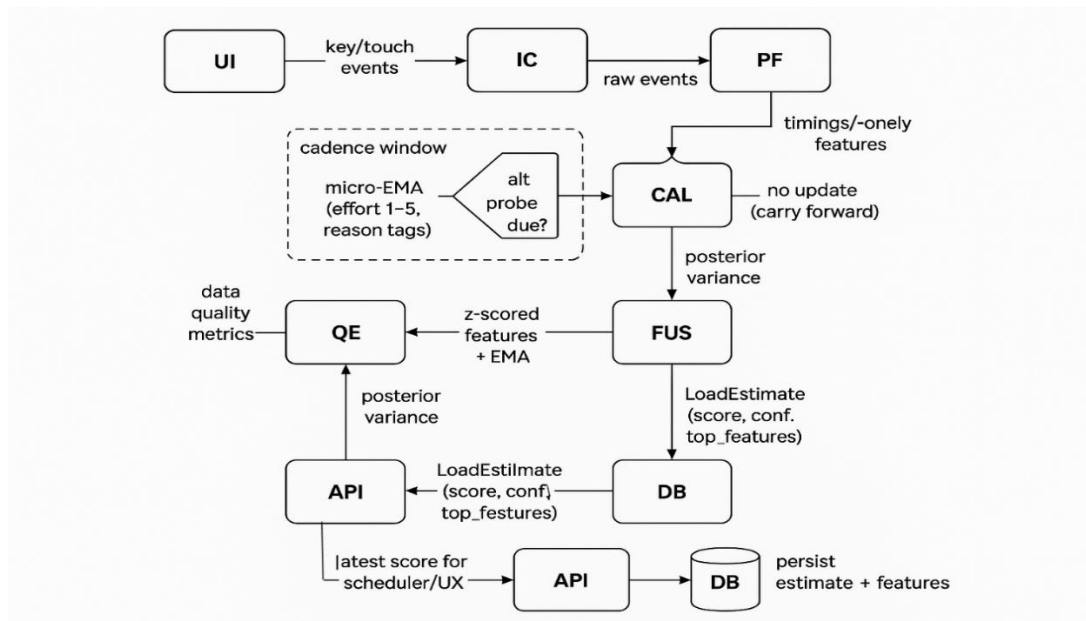
Numerous studies demonstrate that typing patterns reflect mental fatigue and stress. Mental fatigue detection via keystroke dynamics has shown area-under-the-curve scores of about 0.72-0.80 in distinguishing fatigued versus rested states [4]. In real-life office work, prolonged typing led to increases in inter-key intervals and backspace corrections, markers linked to fatigue [1]. Such findings indicate that subtle changes in typing speed, pause burstiness and error corrections can serve as digital biomarkers for cognitive load. By combining passive keystroke/touch metrics with micro-EMA ground truth, it is possible to build a responsive cognitive load estimator.

2.4 Privacy considerations

Passive sensing raises legitimate privacy concerns. However, keystroke dynamics research shows that collecting only timing metadata (press and release times, delays, error corrections) avoids capturing the content of what is typed, protecting user confidentiality [3]. Further, modern apps can process data locally, eliminating the need to send sensitive interaction logs to cloud servers. Ensuring transparency about what is collected and allowing users to opt-in for exporting aggregated, anonymised data is essential for trust.

2.5 Current study tools and gaps

Most commercial study timers implement fixed schedules and ignore cognitive state. They lack mechanisms to detect mental fatigue or adapt to break timing. None combine passive keystroke/touch sensing with micro-EMA to estimate load in real time. Academic work on cognitive load estimation often relies on wearables like EEG or heart-rate sensors, which are intrusive and impractical for everyday study [2]. Research suggests that cognitive load can be measured by less time consuming, less burdensome and unobtrusive methods in real time with decent accuracy. Some of above said methods are facial analysis models using a camera [6], pointer behavior analyzing (mouse only) [7], Micro EMA [8] as well as analyzing keystrokes (keyboard) [9]. Our proposed cognitive load estimator models is expected to have higher accuracy because it leverages both computer interactions and micro-EMA to unobtrusively measure cognitive load during natural study sessions.



UI - StudySession UI, IC – InteractionCapture, PF – PrivacyFilter,
 FX – FeatureExtractor, CAL – Calibrator, FUS – FusionEngine, QE – DataQuality,
 API – EstimatorAPI

Figure 2.1: Conceptual architecture of the dual-signal cognitive load estimator subsystem.

This diagram Figure 2.1 illustrates the workflow of a **cognitive load estimation** system. The process begins when the **StudySession UI** captures a user's **key/touch events**, which are then filtered by the **InteractionCapture** and **PrivacyFilter** to become clean, raw events. The **FeatureExtractor** processes these events to derive timings and one-shot features. The **Calibrator** component periodically asks for user effort feedback to refine its model, a crucial step for personalized accuracy. The **FusionEngine** combines the features with the calibrated data to produce a **LoadEstimate** score and confidence. The **EstimatorAPI** and **Database** work together to store and retrieve these estimates. Finally, the **DataQuality** component continuously assesses the quality of the data, ensuring the system maintains high standards for reliability and accuracy.

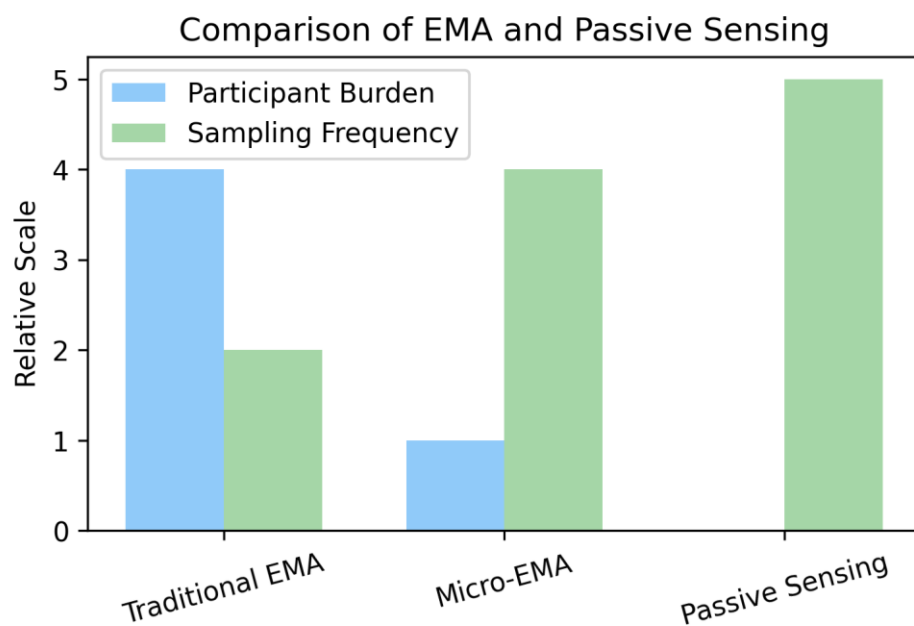


Figure 2.2: Comparison of traditional EMA, micro-EMA and passive sensing on participant burden and sampling frequency (relative scale).

3 Research Gap and Problem Statement

3.1 Research Gap and Problem Statement

Despite the prevalence of digital productivity tools, a significant gap exists in creating study timers that are both intelligent and respectful of user context. Students often rely on timers that assume one schedule suits all, yet cognitive load and fatigue fluctuate significantly. Our literature review identifies four key gaps that this project aims to address:

1. **Static vs. Adaptive Scheduling:** The majority of commercial study timers (e.g., those using the Pomodoro technique) employ static, pre-defined work/break intervals. They lack the mechanism to adapt to a user's real-time cognitive state, potentially interrupting deep focus or failing to provide breaks when fatigue sets in. This ignores research showing that short, well-placed breaks are beneficial, but their optimal timing varies by person and context.
2. **Intrusiveness of Existing Measurement Techniques:** Academic research has validated several methods for cognitive load detection, but they often rely on technologies that are ill-suited for everyday contexts. Traditional methods use intrusive physiological sensors like EEG and ECG or require cameras, which are impractical for daily study sessions and raise privacy concerns [6]. While frequent, multi-item questionnaires like traditional Ecological Momentary Assessment (EMA) can be disruptive and lead to high participant burden, reducing adherence.
3. **Untapped Potential of Single-Modality Interaction Dynamics:** Passive sensing of computer interaction has shown promise. Studies have demonstrated that analyzing mouse activity alone [7], keystroke dynamics alone [9], or using only micro-EMA [8] can correlate with cognitive states. However, these data streams have been primarily explored in isolation or for applications like user authentication, not for powering real-time, adaptive interventions in consumer-facing productivity tools. No existing tool effectively fuses these signals for a more robust estimation.

4. **Lack of Fusion-Based Models for Improved Accuracy:** While micro-EMA offers a lightweight method for gathering subjective feedback, there is limited research on *fusing* these sparse, high-quality subjective reports with continuous but noisy passive interaction data. By combining these signals, it may be possible to create a more accurate and robust personalized cognitive load score, overcoming the limitations of single-modality systems for just-in-time study support.

3.2 Problem Statement

The core problem this research addresses is that **students and learners lack an effective, non-intrusive, and privacy-preserving tool to manage their study sessions according to their naturally fluctuating cognitive load, leading to inefficient learning, frustration, and potential burnout.**

This central problem can be deconstructed into the following key research questions:

1. **Technical Problem:** How can cognitive load be accurately and continuously estimated in real-time using readily available computer interaction sensors (keyboard, mouse, trackpad) without compromising user privacy by capturing sensitive content?
2. **Methodological Problem:** How can sparse but reliable subjective self-reports (via micro-EMA) be effectively fused with continuous but potentially noisy passive sensor data to create a robust and personalized model that outperforms systems relying on a single data stream?
3. **Usability Problem:** How can this sensing technology be implemented within a study application in a manner that is minimally burdensome to the user and provides a tangible benefit to their study habits and well-being?

4 Objectives

To address the identified research problem, this project will pursue a primary objective supported by several specific, measurable objectives.

4.1 Main Objective

To design and evaluate a privacy-first subsystem that unobtrusively estimates cognitive load during study sessions by fusing typing/touch/pointer dynamics (computer interactions) with micro-EMA, providing a real-time load signal to support personalized break scheduling.

4.2 Specific Objectives

1. **To design a dual-signal instrumentation and data collection framework.**
This involves implementing a robust, privacy-preserving system to capture computer interaction cadence (e.g., inter-key intervals, pause distributions, pointer movements) and developing a low-burden micro-EMA prompting mechanism [8] to gather in-situ subjective feedback on cognitive effort. The framework must ensure data quality while remaining unobtrusive.
2. **To implement a novel cognitive load estimator by fusing dual data streams.** The core technical contribution is to design and train a machine learning model that effectively fuses the two data streams (passive interaction patterns and subjective micro-EMA responses). This stands in contrast to previous studies that have focused on single modalities like mouse-only [7], micro-EMA-only [8], or camera-based approaches [6]. The novelty lies in creating a more accurate and reliable real-time cognitive load score through this fusion, using a no-camera, no-wearable approach.
3. **To integrate the cognitive load estimator with an adaptive scheduler.** This objective involves developing a clear API that allows the estimator to provide its real-time score and confidence intervals to a higher-level adaptive scheduling module. This demonstrates the practical application of the sensing subsystem as the foundational component of a complete, intelligent study timer that can personalize interventions, such as break recommendations.

4. **To empirically evaluate the performance, usability, and effectiveness of the estimation model.** A subjective user study will be conducted to assess the model's predictive accuracy against established subjective workload measures (e.g., NASA-TLX). This evaluation will also gather qualitative feedback on user acceptance, perceived burden, and the overall utility of the system in a real-world study context, ensuring the solution is not only technically sound but also user centered.

5 Methodology

5.1 Research Design

The study will adopt mixed methods, within-subject design. Participants ($n \approx 15$) will be people in random education levels (Grade school, High school, University etc.) engaged in self-study tasks over a four-week period. Each participant will install the study timer application on their respective devices. The sensing module will run locally, logging interaction cadence and micro-EMA responses. For comparison, participants will periodically complete a NASA-TLX after study sessions to provide an external subjective workload measure. Semi-structured interviews will be conducted post-study to understand user experience and acceptability.

5.2 Data Collection

5.2.1 Passive interaction features

The app will record timing metadata during study sessions:

- **Inter-keystroke interval (IKI):** time between consecutive keypresses or touches. Increased IKI indicates slower typing and is associated with mental fatigue [1].
- **Burstiness and pauses:** distribution of pauses longer than threshold (e.g., 1 s). Long pauses and erratic timing may signal cognitive overload.

- **Backspace usage:** proportion of backspace events among keypresses; more corrections suggest declining accuracy due to fatigue [1].
- **Touch/Pointer rhythm variability:** standard deviation of IKI and pressure/duration (if available). High variability may reflect lapses or divided attention.

No content of typed text will be stored; only timestamps and event types will be recorded, respecting privacy [3].

Table 5.1: Summary of features used for cognitive load estimation

Feature	Description
Inter-keystroke interval (IKI)	Time between consecutive keypresses; increases as fatigue builds
Burstiness and pauses	Distribution of pauses longer than a threshold; long pauses indicate overload
Backspace usage	Proportion of backspace events; higher rates reflect increasing errors and fatigue
Touch/Pointer rhythm variability	Standard deviation of IKI and pressure/duration; high variability may indicate lapses

5.2.2 Micro-EMA prompts

Micro-EMA questions will ask participants to rate their current mental effort or perceived strain on a 5-point scale. Initial questions include:

- “How mentally taxed do you feel right now?” (0 – not at all to 4 – extremely)
- “How easy or hard is the current study task?”

Prompts will appear every 20–30 minutes or adaptively based on variability in keystroke dynamics. Micro-EMAs will be delivered via a two-tap UI that can be dismissed quickly. The design follows evidence that single-question EMAs reduce burden and increase adherence [2].

5.2.3 Survey and interview data

Participants will complete a NASA-TLX after selected sessions to provide a reference workload measure [5]. At the end of the study, semi-structured interviews will explore perceptions of privacy, intrusiveness and usefulness of the sensing module.

5.3 Feature Extraction and Modelling

Collected raw data will be processed as follows:

1. **Pre-processing:** Remove outliers (e.g., IKI > 5 seconds) to exclude interruptions not related to typing [1]. Align data streams and resample to a common timeline.
2. **Feature engineering:** Compute mean, median and variance of IKI per minute; frequency and duration of pauses; ratio of backspace presses; typing speed trend; and other time-domain features. Extract exponential moving averages to smooth short-term fluctuations.
3. **Label creation:** Micro-EMA responses will be normalized (0–1) to serve as ground truth for perceived load. NASA-TLX subscales will be used to verify construct validity.
4. **Model training:** Evaluate multiple algorithms (linear regression with regularization, random forest, gradient boosting, and Bayesian filters) using k-fold cross-validation. Because cognitive load is subjective and varies between individuals, models will be trained per participant to capture within-person patterns. A hierarchical model could pool data across participants to bootstrap initial estimates then adapt online.
5. **Fusion of signals:** Investigate combining micro-EMA and passive features through Kalman filters or Bayesian updating, weighting each signal according to confidence. Micro-EMA provides sparse but reliable labels, while keystroke/touch dynamics offer continuous but noisy signals.
6. **Model update:** Implement self-learning to adjust the model as more data become available, ensuring the estimator remains accurate over time.

5.4 Requirement specification

This section outlines the user and functional requirements for the Adaptive Cognitive-Load Study Timer. These requirements are derived from the project's objectives and are designed to ensure the final system is effective, usable, and secure.

5.4.1 User requirements

User requirements describe the goals and expectations from the perspective of the end-user, primarily students and knowledge workers.

- **UR1: Effortless Cognitive Awareness:** The user needs a way to understand their own mental state during study sessions without actively self-monitoring. They want the application to "just know" when they are focused, getting tired, or feeling overloaded, so they can make better decisions about their work habits.
- **UR2: Personalized and Timely Break Guidance:** The user wants to be prompted to take breaks at the right time for them, not based on a generic clock. The recommendations should feel intelligent, helping them rest before burnout occurs and return to their tasks refreshed.
- **UR3: Non-Intrusive and Minimalist Experience:** The application must integrate seamlessly into the user's workflow. It should operate quietly in the background, with any interactions (like feedback prompts) being extremely brief, simple, and easy to dismiss so as not to break their concentration.
- **UR4: Absolute Privacy and Trust:** The user must have complete confidence that their privacy is protected. They need assurance that the application is not a keylogger, does not read their screen, and does not send sensitive personal data to the cloud. They want transparency about what data is collected and full control over it.
- **UR5: Simple and Intuitive Interface:** The user needs a clean, straightforward interface to start, pause, and end study sessions. The visual display of information, such as cognitive load trends or break timers, should be easy to understand at a glance.

5.4.2 Functional and Nonfunctional requirements

Functional requirements define the specific behaviors and functions the system must perform to meet the user requirements.

FR 1: Cognitive Load Sensing Module

- FR 1.1: The system shall passively capture keystroke timing metadata, including inter-key intervals, pause frequency and duration, and backspace usage, during an active session.
- FR 1.2: The system shall passively capture pointer and touch interaction dynamics, such as movement velocity, click rate, and scroll behavior.
- FR 1.3: The system shall not record the content of any key presses (e.g., 'A', 'b', '!') or any text content from the screen. Only timing and event-type metadata will be stored.
- FR 1.4: The system shall present a single-question micro-EMA prompt to the user at adaptive intervals to gather a subjective rating of their current mental effort.
- FR 1.5: The micro-EMA interface shall be dismissible with a maximum of two taps.
- FR 1.6: The system shall perform all data processing and model inference for cognitive load estimation on the user's local device.
- FR 1.7: The system shall compute and output a normalized cognitive load score and an associated confidence level in real-time.

FR 2 : System Interface and Data Management

- FR 2.1: The user interface shall provide clear controls to start, stop, and pause a study session.
- FR 2.2: The system shall include a "Transparency Panel" in the settings that clearly explains what data is being collected, for what purpose, and how it is stored.

- FR 2.3: The system shall provide a mechanism for the user to export an anonymized summary of their session data (e.g., load scores, session durations).
- FR 2.4: The system shall securely store all collected metadata and user responses in an encrypted local database.

NFR 1: Non-Functional Requirements

- NFR 1.1 (Performance): The application's background sensing processes must have a minimal impact on system performance and battery life, consuming less than 5% of additional battery capacity during a typical one-hour session compared to baseline usage.
- NFR 1.2 (Privacy): No raw or processed interaction metadata shall be transmitted to any external server or third party without explicit, opt-in consent from the user for research purposes.
- NFR 1.3 (Usability): The system's adaptive algorithm for scheduling micro-EMA prompts must be tuned to avoid prompting the user during periods of high interaction intensity (e.g., fast typing bursts).
- NFR 1.4 (Reliability): The application must remain stable and operational during extended study sessions of up to three hours without crashing or failing to record data.

5.5 Implementation

A cross-platform application will be developed with the following components:

- **Interaction logger:** A custom keyboard and touch listener capturing event timestamps, key types and metadata. The logger will run only during activated study sessions.
- **Micro-EMA scheduler:** Logic to display micro-EMA prompts at adaptive intervals, ensuring prompts do not interrupt the user excessively. The scheduler will track responses and log completion time.

- **On-device processing:** Feature extraction and model inference will occur locally. The app will maintain an encrypted database (e.g., SQLite with Room) storing timing metadata and micro-EMA responses.
- **Transparency panel:** A user interface explaining what data are collected, how load scores are computed, and offering options to export anonymised session summaries. This addresses privacy concerns and fosters trust.
- **API for scheduler:** The sensing module will expose current load score and confidence to the adaptive scheduler. When the load estimate crosses a threshold or trends upward, the scheduler may trigger a break recommendation.

The system will be implemented with energy efficiency in mind, using background tasks and throttling to minimize battery impact.

5.6 Evaluation Plan

Evaluation will focus on accuracy, responsiveness, user burden and overall utility.

1. **Predictive accuracy:** Compute correlation (Pearson's r) and mean absolute error between predicted load scores and micro-EMA responses. Compare models trained with only passive features versus combined signals. Evaluate whether load estimates correlate with NASA-TLX scores and objective performance measures (typing speed, error rates). An acceptable baseline might be $r > 0.5$ and $MAE < 0.7$ on the 5-point scale.
2. **Temporal responsiveness:** Assess how quickly the model detects changes in cognitive load after a change in typing dynamics or micro-EMA rating. Time-to-detect should be under 2 minutes to support adaptive scheduling.
3. **User burden and acceptance:** Use post-study surveys to rate perceived intrusiveness of micro-EMA prompts and satisfaction with the sensing module. Interview transcripts will be coded for themes around ease of use, privacy concerns and perceived benefit.
4. **Battery and performance impact:** Measure CPU and battery consumption of the sensing module during pilot sessions to ensure viability for daily use.

5. **Contribution to scheduling:** Although evaluation of the full adaptive scheduler is beyond this feature’s scope, we will conduct small-scale simulations where break suggestions are triggered by the load estimator. Observing whether users report timely and helpful break suggestions will provide preliminary evidence of utility.

Table 5.2: Evaluation of metrics and targets

Metric	Purpose	Target
Pearson correlation (r)	Measures correlation between predicted load and micro-EMA ratings	$r > 0.5$
Mean absolute error (MAE)	Average absolute difference between predicted and reported load (0–4 scale)	$MAE < 0.7$
Time-to-detect	Time to detect change in load after change in typing dynamics or micro-EMA	< 2 minutes
Battery impact	CPU and battery consumption during sensing	Minimal ($< 5\%$ extra battery)

Ethical approval will be sought from the university’s review board, and informed consent will be obtained from all participants. Data will be anonymised, and participants can withdraw at any time.

6 Description of personal and Facilities

Personal Roles and Expertise

- **Bogahawatta B.P.S. (IoT/Sensing/ML)** – Leads the sensing module, designs the data collection framework, develops the interaction logger, micro-EMA scheduler as well as cognitive load estimator model and manages privacy controls. With unobtrusive micro-EMA and passive computer interaction capturing; high-quality, low-overhead data capture is ensured.

- **Collaborating Team Members:** While this proposal focuses on the sensing component, successful integration relies on collaboration with the AI/ML specialist (who will develop the contextual bandit scheduler), the backend developer (who manages secure local storage and data export) and the UI developer (who implements the user interface and integrates models).
- **Supervisors:** Provide guidance on study design, ethics and evaluation. Access to a human-computer interaction lab and expertise in statistical analysis will support model development.

Facilities and Equipment

- **Devices:** Participants' own devices will be used; a small pool of test devices will be available for those without compatible phones.
- **Software tools:** Flutter/Java/JavaScript, Python (for analysis), version control (Git).
- **Study environment:** Participants will use the app in their normal study settings. A controlled lab environment will be available for initial pilot testing and debugging.

7 Work Breakdown Structure (WBS)

Table 7.1: Work breakdown structure (WBS)

Phase & Task	Duration (weeks)	Deliverables
1. Planning & Literature Review	2	Annotated bibliography on cognitive load, micro-EMA, keystroke dynamics; refined problem statement
2. Instrument & Protocol Design	4	Finalized micro-EMA questions and prompting schedule; interaction logging specification; ethics application draft

Phase & Task	Duration (weeks)	Deliverables
3. App Prototype Development	5	Initial application module capturing keystroke/touch metadata; micro-EMA UI; local storage; transparency panel
4. Pilot Data Collection	4	Pilot study with 3–5 users; preliminary data set; feedback on usability
5. Feature Engineering & Model Prototyping	6	Feature extraction scripts; baseline models trained; evaluation metrics report
6. Full-Scale Study Implementation	6	Recruitment of ~30 participants; deployment of app; ongoing data monitoring
7. Data Analysis & Model Refinement	6	Trained personalized models; statistical analysis of correlation with micro-EMA and NASA-TLX; model selection report
8. User Evaluation & Interviews	4	Survey and interview instruments; coded themes; user burden assessment
9. Integration & API Development	4	API delivering real-time load scores; documentation for scheduler integration
10. Final Report & Dissemination	4	Research paper/report; poster; presentation of findings; recommendations for further development

The overall timeline spans approximately twelve months, overlapping tasks where possible. Regular meetings with supervisors will track progress, and deliverables will be reviewed at each milestone.

8 Self-Evaluation Plan

To ensure quality and rigour, the following self-evaluation criteria will be used:

- **Relevance and completeness of literature review:** Does the survey cover seminal and recent work on cognitive load measurement, EMA and keystroke dynamics? Are the identified gaps clearly articulated?

- **Usability of the sensing module:** Are micro-EMA prompts short enough to minimize interruption? Is the interaction logger stable across devices? Pilot participants' feedback will inform revisions.
- **Model performance:** Do trained models achieve the targeted correlation (>0.5) and low error? Are they robust across participants? Is there evidence of overfitting?
- **Ethical compliance:** Has participant consent been appropriately obtained? Are data stored securely? Is the transparency panel clear about data collection and processing?
- **Integration readiness:** Does the module expose a clean API? Can the adaptive scheduler easily consume load scores? Are response times acceptable (<200 ms)?

Progress will be documented weekly, and adjustments made as needed. External peer review (e.g., supervisor feedback) will supplement self-evaluation.

9 Budget

Budget for the discussed cognitive load estimator module remains very low, because the necessary costs like material (device) cost, hardware cost, software cost are nonexistent. Developers are expected to use their own devices and hardware as well as the software (vs code, postman, python) used in development are free and open source. Libraries used are also free and opensource (e.g.: sklearn) In the development and testing stage, free hosting like vercel is used as there are not any scaling or load handling requirements in the said stage. Only miscellaneous cost like electricity cost, internet cost are expected when strictly considering the project.

10 Commercialization Potential

The proposed sensing module has several attributes that support commercialization. First, it is self-contained and privacy-preserving; it does not rely on cloud services or expensive hardware. Second, many students already use study timers; adding intelligent, state-aware scheduling offers tangible value. The

dual-signal approach could be packaged as a stand-alone SDK for other productivity apps, expanding reach beyond our timer. Third, growing awareness of digital well-being means there is a market for tools that help people manage focus without intrusive surveillance. Nonetheless, further work is needed to demonstrate scalability, battery efficiency and long-term adherence before commercial deployment. Partnering with educational institutions or productivity app developers could accelerate adoption.

11 Conclusion

This proposal outlines a research plan to develop and evaluate a dual-signal sensing subsystem that combines passive typing/touch dynamics with micro-EMA to estimate cognitive load during study sessions. By synthesising insights from cognitive psychology, human–computer interaction and machine learning, the project addresses the limitations of fixed-schedule study timers. The literature shows that typing performance deteriorates as mental fatigue builds [1] and that micro-EMA provides a low-burden means to capture momentary perceptions [2]. By fusing these signals in a privacy-first application, we aim to deliver a responsive and personalized cognitive load estimator with improved accuracy. The proposed methodology covers instrument design, data collection, modelling and evaluation, ensuring both scientific rigor and user acceptability. Successful completion will not only contribute a novel feature to the adaptive study timer but also advance knowledge in unobtrusive cognitive load monitoring.

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